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Influence of composition on the structure and conductivity of the fast ionic conductors $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ ($0.03 \leq x \leq 0.167$)

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Abstract

The solid solution $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ ($0.03 < x < 0.167$) has been investigated by powder X-ray diffraction (XRD), impedance spectroscopy (IS) and ^7Li nuclear magnetic resonance (NMR) techniques. In these samples, a change of symmetry from tetragonal to orthorhombic is observed when the lithium content is decreased below $x = 0.06$. Structural modifications produced are mainly due to cation vacancies ordering along the c -axis, which disappear gradually when the Li content increases. Two Li signals with different quadrupole constants are detected in ^7Li NMR spectra of orthorhombic/tetragonal perovskites, which have been associated with two crystallographic sites. In $\text{La}_{0.5}\text{Li}_{0.5}\text{TiO}_3$ perovskite, prepared by quenching from 1300°C into liquid nitrogen, an important mobility for Li was detected in ^7Li NMR spectra. In perovskites analysed, the dependence of electrical conductivity on Li content departs from that expected on the basis of a random distribution of La and Li in A sites. Dc conductivity increases quickly with the Li content in orthorhombic samples, but changes much more smoothly in tetragonal ones, where vacancies distribution becomes progressively disordered. In all perovskites analysed, the dependence of dc-conductivity with temperature displays a non-Arrhenius behaviour, with activation energies of 0.39 ± 0.02 and 0.29 ± 0.02 eV in 160–250 and 250–360 K temperature ranges respectively. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: ^7Li NMR; Ionic conductivity; Perovskite structure; Lithium mobility

1. Introduction

Recently, a wide variety of solid electrolytes with high ionic conductivity have been prepared with potential application as solid electrolytes in high energy batteries and other electrochemical devices

[1]. Since Inaguma's group reported in 1993 the outstanding electrical properties of lithium lanthanum perovskites [2], numerous works have been oriented to relate the crystal chemistry to ionic conductivity in these perovskites. In particular, the solution of rare earth (RE) and Li cations in perovskites (ABO_3), produced materials with formula $\text{RE}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ [3–5]. In this system, the maximum conductivity was found for La series ($10^{-3} \text{ S cm}^{-1}$ at room temperature). Partial substitution of La

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by bigger cations such as Sr, Ba and Na in A sites of perovskites produced variable effects [6]. In the case of Sr, the ionic conductivity increased slightly for low substitution degrees, in that of Ba no single phases were obtained and in that of Na, the conductivity decreased quickly with dilution [7]. Substitutions of Ti (located in B sites of perovskites) by other tetravalent cations, such as Zr, Sn, Mn or Ge [8], showed that ionic conductivity decreased when the ionic radii increased. The replacement of Ti by pentavalent cations like Nb [9,10] or Ta [11] decreased again the ionic conductivity.

All mentioned compounds have the perovskite structure; however, structural distortions produced in each case have not been determined. In the case of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ compounds, a tetragonal distortion is detected along the series, with c/a ratios depending on the Li content of the perovskite [4,12]. In La-rich samples, La and vacant A sites alternate along the c -axis in contiguous planes, however vacancies become progressively disordered when the Li content increases. Electrical conductivity of these perovskites increases with Li content and for several compositions analysed, it depends on quenching temperatures [5,13], indicating that modifications on the microstructure of these materials could also affect the conductivity.

In this work, the influence of the composition on the structure of the $\text{Li}_{3x}\text{La}_{2/3-x}\text{TiO}_3$ series has been studied. XRD patterns of perovskites with different Li contents have been analysed by the Rietveld method, and structural features correlated with those obtained by ^7Li NMR spectroscopy. Finally, electrical conductivity data have been interpreted on the basis of structural information deduced.

2. Experimental

Samples were prepared by conventional solid-state reaction. The reagents used were Li_2CO_3 (Merck), La_2O_3 (Aldrich 99.99%) and TiO_2 (Aldrich 99%). La_2O_3 was heated at 700°C prior to weighing. Stoichiometric amounts of these reagents were ground together in an agate mortar and heated at 800°C for 4 h in order to eliminate CO_2 . The reground products were cold-pressed at 150 MPa and heated at 1150°C for 12 h. Finally, powders were pressed and heated again at 1350°C for 6 h. In order

to avoid lithium losses the heating rate used in all treatments was 1°/min.

X-ray diffraction powder (XRD) patterns were recorded with $\text{Cu K}\alpha$ radiation in an X'Pert Philips diffractometer, with $(\theta/2\theta)$ Bragg–Brentano geometry, equipped with a curved graphite monochromator. Data were taken with a 0.5° divergence slit, a receiving slit of 0.01° and a set of soller slits with an axial divergence of 1°. The 2θ range analysed was 10–90°, with a step scan of 0.02 and a counting time of 10 s for step. In a first stage, XRD patterns were indexed with the Dicvol program [14]; afterwards, patterns were analysed by the Rietveld method, using the GSAS program [15].

^7Li NMR spectra were taken in a MSL 400 Bruker spectrometer working at 155 MHz. Static spectra were taken after irradiation of the sample with a $\pi/2$ pulse (3 μs). The recycling time used between accumulations was 10 s and the number of scans 100. Filter used in signal detection was 100 kHz. Chemical shifts of NMR signals were referred to that of 1 M LiCl aqueous solution. The fitting of the experimental patterns was carried out using the WINFIT Bruker Program. Intensity, position and linewidth of components were determined with a non-linear iterative least-squares method. However, quadrupole C_Q and η constants were determined by a trial and error procedure.

Electrical conductivity measurements were carried out by complex Impedance Spectroscopy using automatically controlled HP 4284A and HP4285A precision LCR meters. Frequency and temperature ranges used were 20 Hz–30 MHz and 150–500 K, respectively. All measurements were performed on cylindrical pellets (5 mm in diameter and 0.7 mm thick) with evaporated gold electrodes, and under a dry N_2 flow to ensure an inert atmosphere. Dc-conductivity values were obtained from the ‘plateau’ observed in conductivity versus frequency plots at frequencies where blocking effects are not present.

3. Results

3.1. X-ray diffraction experiments

Fig. 1 shows the powder X-ray diffraction (XRD) patterns, recorded at room temperature, of perovskites $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$, with $x=0.03$, $x=0.063$, $x=$

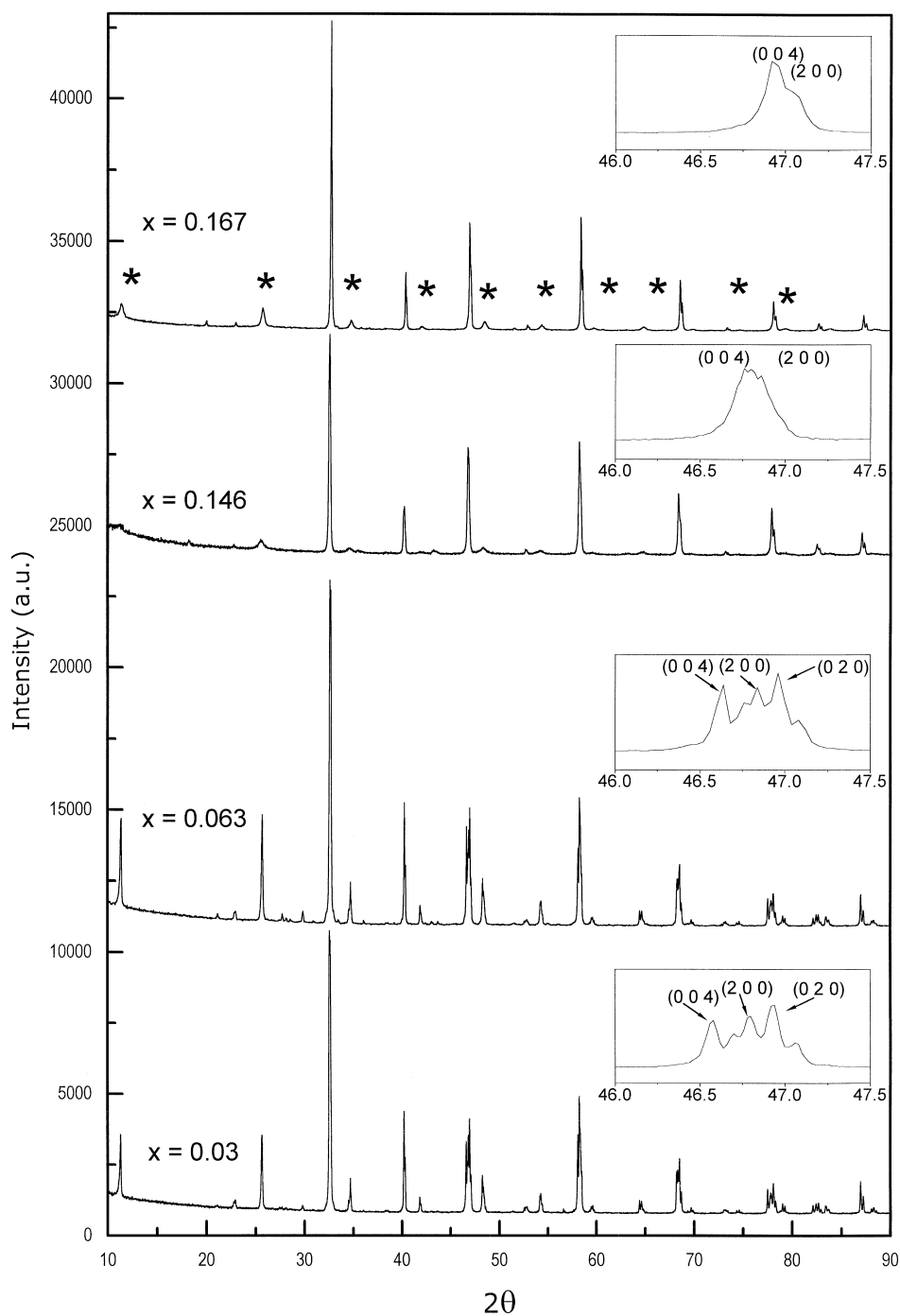


Fig. 1. Powder X-ray diffraction patterns of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites for four different compositions: $x = 0.03$, $x = 0.063$, $x = 0.146$ and $x = 0.167$. Superstructure peaks corresponding to the doubled perovskite are labelled with asterisks in the X-ray pattern of the $x = 0.167$ perovskite. Splitting of the (200) reflections are shown as insets.

0.146 and $x=0.167$. Two different patterns can be obtained depending on the lithium content of these perovskites.

For samples with high lithium contents ($x=0.146$ and 0.167), XRD patterns could be indexed with the diagonal perovskite cell ($\sqrt{2}a_p \times \sqrt{2}a_p \times 2a_p$) deduced in a previous microstructural study [16]. However, the average crystal structure deduced from XRD experiments could be interpreted with a smaller cell. The best fitting of the XRD patterns was achieved using a primitive $P4/mmm$ tetragonal cell, derived from the cubic perovskite, with $a \approx a_p \approx 3.87$ Å and $c \approx 2a_p$ [2,4,12]. In these samples, peaks associated with the doubled cell, labelled with asterisks in Fig. 1, are broad while those of the cubic sublattice are narrower.

For samples with low lithium content, $x < 0.0625$ the XRD patterns are slightly different. Splittings of different peaks have been detected, in particular, $\{200\}$ lines became into three components (see inset of the Fig. 1), indicating the presence of an orthorhombic distortion in the perovskite network. All peaks of the XRD patterns were indexed with the $Pmmm$ space group, and the unit cell parameters

were $a \approx a_p \approx 3.864$ Å, $b \approx a_p \approx 3.875$ Å, $c \approx 2a_p \approx 7.786$ Å [17]. Note that, in the case of orthorhombic perovskites, peaks of the superstructure are sharper and stronger than those detected in tetragonal phases. In the case of tetragonal samples with $x=0.167$, a fast cooling from 1350°C produced an XRD pattern without superstructure peaks (Fig. 2) that could be indexed with a cubic cell ($a \approx 3.87$ Å).

3.2. ^7Li NMR experiments

^7Li NMR spectra of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites, recorded at room temperature are shown in Fig. 3. The spectra are formed by an important central ($-1/2, 1/2$) transition and two broad and small ($\pm 1/2, \pm 3/2$) components, which are symmetrically disposed with respect to the central one.

A deeper analysis of NMR spectra of orthorhombic samples ($x < 0.06$) shows the existence of two Li NMR components, with different quadrupole constants ($C_Q \approx 80$ and 0 kHz), whose relative intensities agree with those deduced from the structural analysis [17]. In tetragonal samples ($x > 0.1$), quadrupole constants decrease considerably ($C_Q \approx 40$ and 0 kHz)

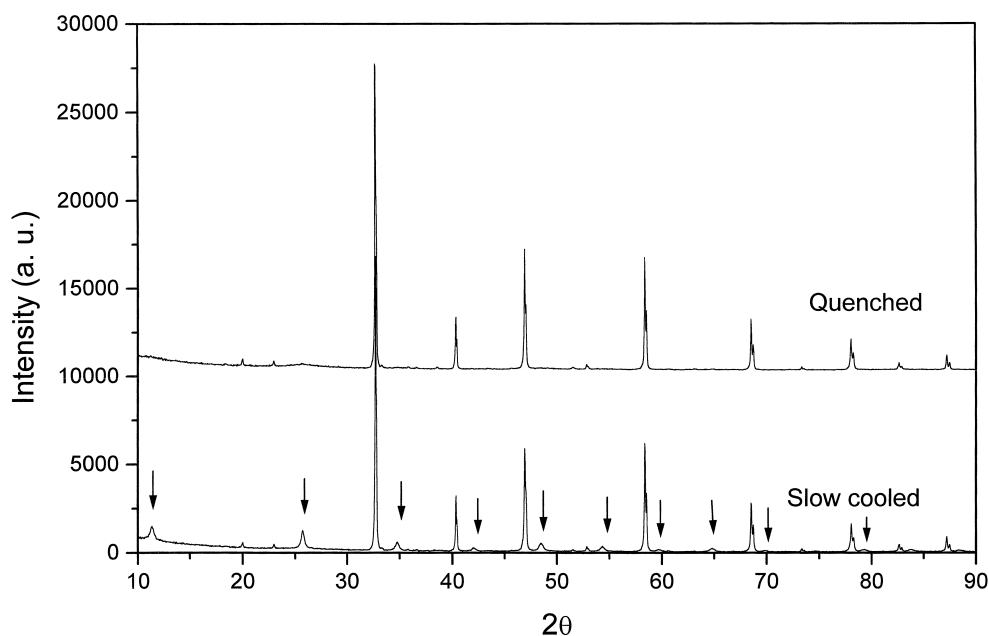


Fig. 2. XRD patterns of the $\text{La}_{0.5}\text{Li}_{0.5}\text{TiO}_3$ perovskite, (a) quenched and (b) slowly cooled from 1350°C . Arrows denote superstructure peaks of the perovskite.

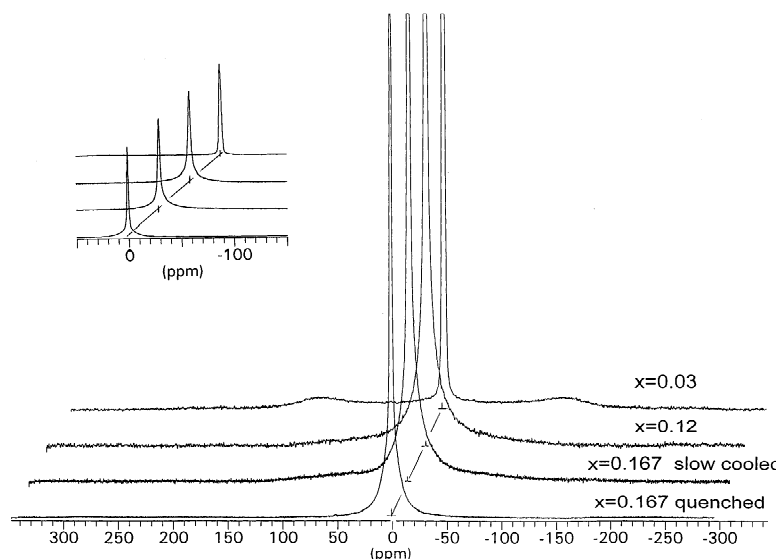


Fig. 3. ^7Li NMR spectra of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites, recorded at room temperature. Quadrupole outer transitions, labelled with asterisks, are only clearly visible in orthorhombic samples. In tetragonal samples outer transition are responsible for the broad component detected at the resonance frequency.

and resolution of two components became difficult because of Li exchange processes. Finally, in tetragonal samples quenched from 1350°C , quadrupole patterns were almost undetectable ($C_Q \approx 0$ kHz).

The central component of the ^7Li MAS NMR spectra displays a gaussian shape in orthorhombic samples but a lorentzian shape in tetragonal ones, indicating that the lithium mobility increases. Finally, in tetragonal samples quenched from 1350°C , the width of the central line decreases indicating an additional increase of Li mobility (see insets of Fig. 3).

3.3. Electrical conductivity

Fig. 4 shows dc-conductivity values measured at 300 K, as a function of lithium content in the solid solution $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$. It can be seen that dc-conductivity increases quickly with lithium content in orthorhombic samples and displays a much weaker dependence in tetragonal ones. The maximum value for dc-conductivity is obtained for lithium content of $x \approx 0.13$.

Temperature dependence of dc-conductivity is shown in Fig. 4 as an inset. It can be observed that all samples show similar temperature dependence.

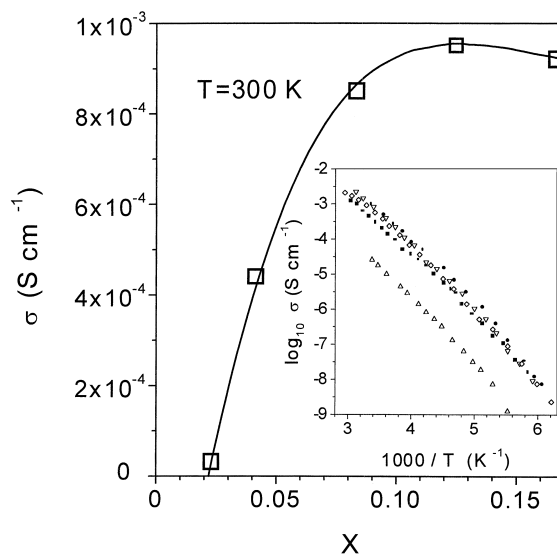


Fig. 4. Room temperature dc-conductivity values versus lithium content, x , of the perovskite. The solid line is plotted as a guide for the eye. The inset shows the temperature dependence for samples with different lithium content: (Δ) $x=0.025$, ($\square$) $x=0.042$, ($\diamond$) $x=0.083$, ($\circ$) $x=0.125$ and (∇) $x=0.167$.

Dc-conductivity is thermally activated, but shows a non-Arrhenius behaviour; activation energies decreases from 0.39 ± 0.2 eV in the low temperatures

(160–250 K) to 0.29 ± 0.02 eV in the high temperature regime (250–360 K). This behaviour is observed in all samples indicating an small dependence of this behaviour on the lithium content of samples.

4. Discussion

The XRD patterns indexing of samples $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$, $0.04 < x < 0.17$, has shown that the unit cell symmetry changes from orthorhombic to tetragonal as Li content is increased. In Fig. 5, dependence of the lattice parameters on the Li content, x , of perovskites is given. A drop of the

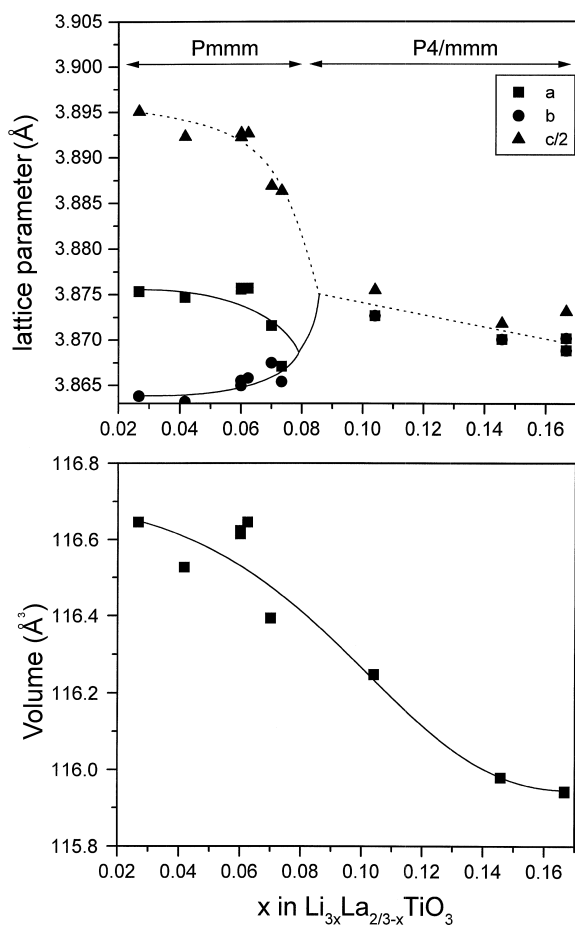


Fig. 5. Plot of the lattice parameters (top) and unit cell volume (bottom) versus Li content in $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites. Uncertainties on unit cell parameters are given in Table 2.

c -parameter is observed for $x > \approx 0.06$. In orthorhombic to tetragonal transformation, a and b parameters become similar and unit cell volume decreases. For high lithium contents, the $c/2a$ ratio reaches values close to 1, and XRD pattern can be described with a cubic cell. Similar observations have been reported in the $(1-x)\text{CaTiO}_3-x\text{La}_{2/3}\text{TiO}_3$ solid solution [18].

At present, the crystal chemistry of lithium lanthanum titanates has not been established. In this sense, different groups [5,12,19] have found similar results in the compositional range $0.04 < x < 0.15$; however, results out of this range are scarce. In these studies, XRD patterns were interpreted on the basis of a tetragonal cell (space group $P4/mmm$), and in the case of Robertson et al. [5], two tetragonal cells, with different c/a ratio, were obtained for high and low lithium contents. Last authors reported also the existence of a cubic phase in the high temperature domain of the ' $\text{Li}_{0.5}\text{La}_{0.5}\text{TiO}_3$ '–' $\text{La}_{2/3}\text{TiO}_3$ ' phase diagram.

In this study the splitting of $\{200\}$ lines has been associated with a orthorhombic distortion of the perovskite. This distortion has been previously reported in oxygen-deficient perovskites $\text{La}_{2/3}\text{TiO}_{3-\delta}$, $0.07 \leq \delta \leq 0.079$, by Abe and Uchino [20], Kim et al. [21] and MacEachern et al. [22]. These authors concluded that the observed double unit cell was produced by the ordered arrangement of La and vacancies along c -axis. The same type of La arrangement has been proposed for tetragonal $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites [5,12,17,19]; however, the orthorhombic splitting has been never detected, and is related to an additional distortion of the octahedral network.

In a second stage of this study, XRD patterns were analysed by means of the Rietveld method. In each sample, a structural refinement was done using the two proposed $P4/mmm$ and $Pmmm$ structural models [11,17,21] (Table 1). In each case, the refinement with the best agreement factors was retained. In this analysis, lithium contribution to XRD patterns was not considered due to its low scattering factor; La contents deduced from Rietveld analysis coincides within 10% with nominal ones. Structural parameters obtained from this analysis are given in Table 2. La ions exhibit a clear preference for (0, 0, 0) with respect to (0, 0, 1/2) positions, but this preference

Table 1

Positional parameters for tetragonal and orthorhombic perovskites used in the structural refinement (Li has not been considered in the refinement)

Atom	Space group $P4/mmm^a$		Space group $Pmmm^b$	
	Position	Atomic coordinates (x,y,z)	Position	Atomic coordinates (x,y,z)
La1	1a	0, 0, 0	1a	0, 0, 0
La2	1b	0, 0, 1/2	1c	0, 0, 1/2
Ti	2h	1/2, 1/2, z	2t	1/2, 1/2, z $\approx$ 1/4
O1	1c	1/2, 1/2, 0	1f	1/2, 1/2, 0
O2	1d	1/2, 1/2, 1/2	1h	1/2, 1/2, 1/2
O3	4i	0, 1/2, z	2r	0, 1/2, z $\approx$ 1/4
O4	–	–	2s	1/2, 1/2, z $\approx$ 4

^a Taken from Ref. [12] for $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ system.

^b Taken from Ref. [21] for $\text{La}_{2/3}\text{TiO}_{3-\delta}$ system.

decreases as the Li content increases (Fig. 6a). La ions distribution becomes completely disordered in tetragonal perovskites with Li content near to $x=0.167$. In this case, samples quenched from 1350°C display a pseudocubic (space group $Pm\bar{3}m$) symmetry and La and Li ions are randomly accommodated in a single site. Samples studied by Fourquet et al. [12] displayed the same trend; however, these au-

thors did not study the full disordered pole because of the narrower compositional range analysed.

In $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ series, titanium ions are displaced from the centre of polyhedra towards $z=0.5$ layer, as deduced from the analysis of Ti–O interatomic distances (Fig. 6b). This fact can be explained on the basis of an asymmetric distribution of charge around the octahedra, produced by the alternating disposition of La and vacancies in contiguous planes of the perovskite. From this fact, one of the Ti–O distances becomes shorter while other ones get longer along the c -axis, producing the mentioned distortion of the TiO_6 octahedra. This distortion decreases when La ions become disordered. In the case of perovskites with $x=0.167$, La ions distribution is completely disordered and octahedra are almost regular.

^7Li NMR spectra of the orthorhombic ($x<0.06$) and tetragonal ($x>0.08$) samples exhibit small but appreciable differences. Structural distortions produced during cation ordering increase the electric field gradient at sites occupied by Li ions favouring detection of the quadrupole pattern. In orthorhombic perovskites, two components with quadrupole constants $C_Q=80$ and 0 kHz were detected. The intense

Table 2

Atomic parameters deduced from X-Ray powder diffraction data of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites^a

	x										
	0.03	0.04	0.06	0.06	0.062	0.07	0.073	0.104	0.146	0.167	0.167
a	3.8753(1)	3.8747(1)	3.8756(1)	3.8757(1)	3.8757(1)	3.8716(1)	3.8671(5)	3.8727(3)	3.8700(1)	3.8688(4)	3.8703(4)
b	3.8638(1)	3.8632(1)	3.8649(1)	3.8655(1)	3.8658(1)	3.8675(1)	3.8654(5)	3.8727(3)	3.8700(1)	3.86885(4)	3.8703(4)
c	7.7901(2)	7.7846(1)	7.7852(1)	7.7845(2)	7.7853(2)	7.7734(2)	7.7727(4)	7.7510(3)	7.7452(2)	7.7463(2)	3.8703(4)
z	2	2	2	2	2	2	2	2	2	2	1
La1(occ.)	0.96(1)	0.90(1)	0.91(1)	0.95(1)	0.91(1)	0.89(1)	0.90(1)	0.75(1)	0.62(1)	0.53(1)	0.5
La2(occ.)	0.32(1)	0.35(1)	0.33(1)	0.29(1)	0.30(1)	0.31(1)	0.29(1)	0.38(1)	0.38(1)	0.47(1)	0.5
Ti z	0.2645(4)	0.2627(4)	0.2637(3)	0.2644(4)	0.2633(5)	0.2611(4)	0.2654(5)	0.2637(5)	0.2584(5)	0.2529(4)	–
O3 z	0.2640(6)	0.2596(6)	0.2610(4)	0.2520(5)	0.2520(4)	0.2810(4)	0.2815(6)	0.2503(7)	0.2497(6)	0.2394(6)	–
O4 z	0.2240(6)	0.2359(5)	0.2450(4)	0.2310(6)	0.2400(4)	0.2290(5)	0.2187(6)	–	–	–	–
dTi–O1	2.060(3)	2.045(3)	2.052(3)	2.058(3)	2.050(4)	2.029(4)	2.080(9)	2.044(6)	2.000(7)	1.959(1)	1.935(1)
dTi–O2	1.835(3)	1.847(3)	1.841(1)	1.835(3)	1.842(4)	1.858(4)	1.807(9)	1.842(4)	1.872(7)	1.914(1)	–
dTi–O3	1.938(1)	1.932(1)	1.933(1)	1.934(1)	1.940(1)	1.940(2)	1.940(1)	1.940(1)	1.9363(5)	1.937(1)	–
dTi–O4	1.956(3)	1.949(2)	1.943(1)	1.957(3)	1.941(2)	1.952(2)	1.941(2)	–	–	–	–
R_p	8.8	10.2	15.7	9.1	9.3	9.9	12.3	11.6	12.6	11.1	15.0
R_{wp}	11.3	14.08	20.9	11.6	13.1	12.8	15.8	16.0	17.9	14.4	19.2
χ^2	1.4	5.07	1.5	1.4	6.9	1.6	1.1	8.3	5.7	2.6	8.1
R_B	6.4	7.6	12.9	8.9	7.4	7.7	9.0	7.0	13.6	11.3	7.2

^a R_p , R_{wp} , χ^2 and R_B are the conventional agreement factors given by the Refinement Program.

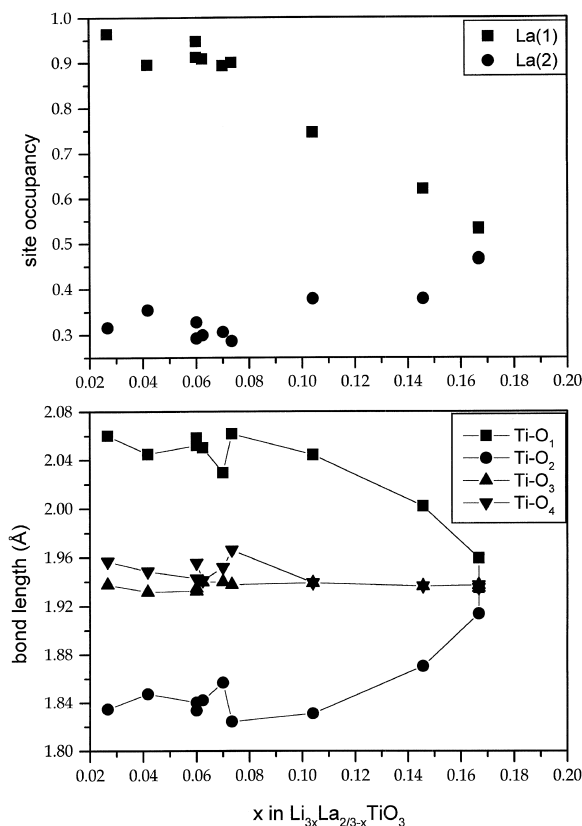


Fig. 6. (top) La occupancies of two structural sites in $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ perovskites. (bottom) Variation of octahedral Ti-O distances with the lithium content of the perovskite. Uncertainties on occupancies and distances are given in Table 2.

narrow line detected near the resonance frequencies corresponds to Li atoms with an important degree of mobility ($C_Q \approx 0$), while the broad ones correspond to Li with a lower mobility ($C_Q \approx 80$ kHz) [17]. Relative intensities of the two components agrees with those deduced from the structural analysis of these perovskites: the broad component has been associated with Li located at $z/c=0$ (30%) and the narrow line to Li located at $z/c=0.5$ planes (70%), where vacancies are preferentially located. In tetragonal perovskites, a small broad non-resolved pattern was detected near the central transitions with $C_Q \leq 40$ kHz. In samples quenched from 1350°C , with $x \geq 0.1$, the intensity and the splitting of the quadrupole pattern decrease, being difficult its detection in ^7Li NMR spectra. In the last samples, a consider-

able narrowing of the central line is produced, indicating that mobility of Li is important.

With regard to electrical conductivity, dc-conductivity depends on lanthanum and vacancies distribution. It is generally assumed that ionic transport in these materials results from lithium ions hopping to near vacant A sites [2,3,23,24]. Therefore, two factors mainly affect the ionic conductivity: the carrier concentration and the probability of ions hopping to neighbor vacant sites. In a first approximation, the effective carrier concentration should be equal to the product $n_c n_v$, where n_c is the lithium and n_v is the vacant A sites concentrations. NMR spin-lattice relaxation and electrical conductivity relaxation rates measured on end members of these perovskites showed that all lithium participates in the ionic ac-conductivity [23–25]. However, some discrepancies between predicted and observed values have been detected that makes the proposed model inadequate. In particular, a random distribution of Li and vacancies on A sites would give conductivity values near zero for $x=0$ ($n_c=0$) and $x=0.167$ ($n_v=0$). In orthorhombic samples, conductivity values increase with n_c in the predicted way. However in tetragonal samples, with high Li contents, conductivity values are always important, suggesting that the amount of vacancies implicated in conductivity must be much higher than considered in this model. This hypothesis has been confirmed in a recent neutron diffraction study of $\text{La}_{0.5}\text{Li}_{0.5}\text{TiO}_3$ perovskite, where Li ions have been located at the centre of the cube faces of the primitive perovskite (1/6 occupancy), instead of the A site [26]. In the resulting rhombohedral $R3c$ perovskite, the amount of vacancies to which Li can hop is clearly higher than assumed in previous works.

The accommodation of cation vacancies in alternating planes along the $\{001\}$ axis should favour a two-dimensional movement of Li in orthorhombic samples [17,19]. However, in tetragonal samples, cation and vacancies turn progressively disordered favouring a three-dimensional movement of Li. Associated with vacancies disordering, a considerable narrowing of the central line is produced, indicating a substantial increment on the mobility of Li along the series. All these facts explain the trend observed on the ionic conductivity of perovskites, in which ionic conductivity increases with the lithium

content, up to the maximum detected at $x \approx 0.13$ in tetragonal perovskites.

In a previous study, it was shown that, successive high temperatures quenching and annealing treatments changed vacancies distribution in tetragonal perovskites. These treatments were responsible for an enhancement of ionic conductivity, by a factor about 2, on quenched samples respect to annealed ones [13,27]. All described observations point to that ionic conductivity of the series $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ depends also on the vacancies distribution.

On the other hand, it has also been observed that structural parameters change only slightly with the Li content. From this fact, it would be expected that the necessary energy by lithium ions to hop into a near A site, should remain almost unchanged. This point has been experimentally confirmed in perovskites with different symmetry, where the same temperature dependence of conductivity has been observed, and activation energies deduced for Li motion changes only slightly along the series. In order to explain the behaviour of these perovskites, additional work must be done.

5. Conclusions

Structure and transport properties of the $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ ($0.03 \leq x \leq 0.167$) solid solution have been investigated. Most of analysed perovskites exhibit a double unit cell ($a_p \times a_p \times 2a_p$), the symmetry changing from orthorhombic to tetragonal when the Li content increases above $x = 0.06$. In Li poor samples, La ions and vacancies alternate in contiguous planes along the c -axis. Distribution of vacancies becomes progressively disordered when the Li content increases, being completely disordered in Li-rich samples quenched from high temperatures (1350°C).

In the NMR study of $\text{La}_{2/3-x}\text{Li}_{3x}\text{TiO}_3$ samples, two types of perovskites have been distinguished in accordance with the XRD study. In orthorhombic perovskites, two types of Li, with different mobility, have been ascribed to Li located in two planes of the perovskite. Li with low mobility, located at $z/c = 0$ planes, display important quadrupole constants ($C_Q = 0.80$ kHz) and Li with a high mobility, located at $z/c = 0.5$ planes, display small constants ($C_Q = 0$ kHz). In tetragonal and cubic phases, quadrupole

constants decrease and resolution of two sites becomes more difficult. The increase of Li mobility along the series is the responsible for the progressive line narrowing of the central transition detected in ^7Li spectra.

Electrical conductivity increases with lithium content in orthorhombic samples but displays only a weak dependence in tetragonal ones. High values of conductivity measured in samples with $x > 0.13$ require a higher amount of vacancies that considered in the random distribution model of La and vacancies. This fact was recently confirmed by neutron diffraction experiments on the $\text{La}_{0.5}\text{Li}_{0.5}\text{TiO}_3$ end member, where an unusual position for Li was detected [26]. In all samples analysed, a non-Arrhenius dependence of dc-conductivity with temperature was obtained.

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